

LLD Case Study 5: Multi-Floor Parking Lot System

Target Patterns: Singleton Pattern, Factory Pattern, Strategy Pattern

Key Engineering Focus: Multi-level spatial modeling, spot allocation algorithms, dynamic fee calculation, and thread-safe entry/exit gates.

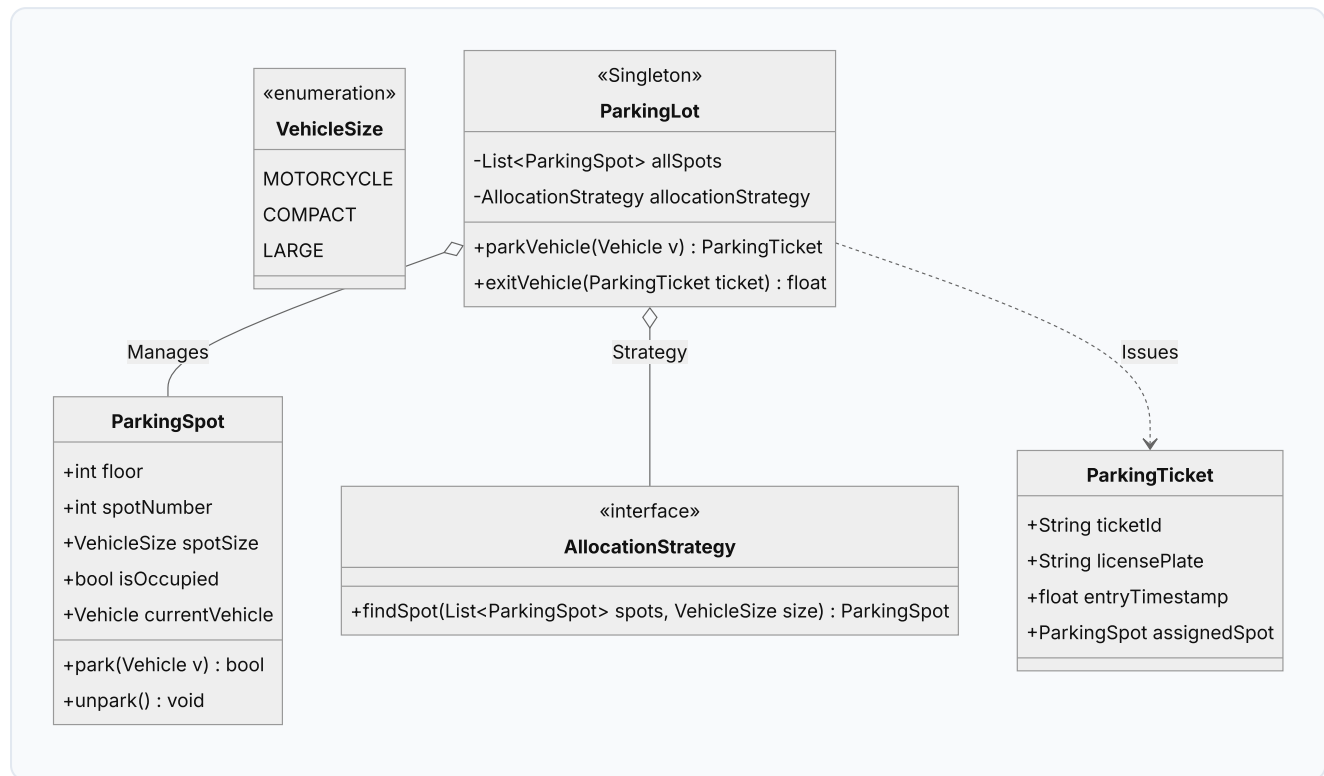
1. Problem Statement & Functional Requirements

Design an automated multi-floor commercial parking lot management system.

Requirements:

1. **Singleton Architecture:** The physical parking lot configuration is a system-wide Singleton.
 2. **Vehicle & Spot Hierarchies (Factory Pattern):** Supports Motorcycles, Cars, and Heavy Trucks fitting into Motorcycle, Compact, and Large spots respectively.
 3. **Spot Allocation Policy (Strategy Pattern):** Pluggable algorithms for spot assignment (e.g. Nearest to Entrance, Lowest Floor First).
 4. **Automated Ticketing & Billing (Strategy Pattern):** Issue timestamped tickets at entry gates and compute fees at exit based on duration and vehicle size multipliers.
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2. Architecture & UML Class Diagram



3. Production-Grade Python Implementation